



MATERIAL POSSIBILITIES

EPP & engineered materials

Product-family catalogue and practical material guide

4 industries. 47 product families.

Explore the current product range, understand EPP behaviour and prepare a clearer specification for your next component or packaging project.

01 Sports, Leisure & Early Childhood

14 product families

02 Logistics & Material Handling

13 product families

03 HVAC

10 product families

04 Automotive

10 product families

The range includes EPP, EPS and related polymers as indicated. Select the material and validate the finished part for its intended application.

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MATERIAL GUIDE

What EPP makes possible

Expanded polypropylene, or EPP, is a thermoplastic bead foam with a predominantly closed-cell structure.

Air-filled beads form components combining low weight, cushioning and insulation. Grade, density and geometry determine performance. [1]

Low weight

A cellular structure reduces the amount of solid polymer in a component. Balance the weight target against support, stiffness and handling loads. [1]

Impact absorption & recovery

EPP can absorb impact energy and recover after deformation. Repeated-impact performance depends on the grade, geometry and severity of loading. [1]

Thermal insulation

The foam structure helps slow heat transfer through a part. Evaluate thickness, joints and operating temperatures in the complete assembly. [2]

Low water absorption

Closed cells limit water uptake within the material. Seams, openings and cleaning conditions still need attention in a finished product. [1]

Noise & vibration control

EPP can contribute to quieter technical assemblies. Request measurements for the actual frequencies, mountings and material grade; acoustic functions differ. [2]

Integrated moulded details

Cavities, ribs and interfaces can be designed into a foam component. Discuss bead access, wall thickness, draft, vent locations and demoulding before fixing the drawing. [5]

PROCESS & SPECIFICATION

From beads to a useful part

01 Prepare & fill

Select expanded beads and fill the mould cavity. Pressure filling introduces beads into a closed tool. [4]

02 Fuse with steam

Steam heats the packed beads so their surfaces bond into the required foam shape. [4]

03 Cool & release

Cool and stabilise the part, then open the mould and eject it with support. [4]

04 Condition & inspect

Apply any specified conditioning, then check dimensions, density, appearance and fit. [4]

Build the application brief

Drawing & interfaces	Share dimensions, contact surfaces, clearances, attachment points and inspection datums. [5]
Loads & impacts	State payload, drop conditions, compression, repeated handling and acceptable deformation. [1]
Operating environment	Describe temperatures, exposure duration, moisture, vibration and cleaning chemicals. [2]
Grade & moulded density	Specify the finished density and tolerance; request the matching grade data. [3]
Application qualifications	Identify food contact, fire performance, electrical protection or UV requirements early. [3]
Tooling & finish	Agree wall transitions, draft, visible vent marks, colour and surface expectations. [5]
Samples & acceptance	Set tests, conditioning, dimensional tolerances and pass criteria before approving production. [5]
Volume & service life	Share quantities, planned reuse, cleaning, collection and the intended recycling route. [4]

INDUSTRY 01 | 14 PRODUCT FAMILIES

Sports, Leisure & Early Childhood

Shape lightweight equipment, play forms and accessories around the people who use them.

Specification focus: Define the activity, age group, anticipated loads and cleaning conditions. Evaluate the complete design against its intended use and relevant safety requirements.

Sports accessories

Moulded polymer and foam accessories for sport-specific equipment concepts and component development.

Material options: EPP / EPS

Custom sports equipment

Custom-shaped sport and leisure components developed around user interaction and the intended equipment function.

Material options: EPP / EPS

Leisure equipment components

Lightweight moulded parts for outdoor, recreation and leisure assemblies.

Material options: EPP / EPS

Foam exercise jump boxes

Lightweight foam exercise-box concepts developed around training use, stability and repeated loading.

Material options: EPP

Sports helmet foam liners

Liner forms developed within complete protective helmets.

Material options: EPS / EPP

Foam insole components

Footwear support forms specified around fit and repeated loading.

Material options: EPS / EPP

Surfboard foam cores

Core geometry developed alongside surfboard construction.

Material options: EPS / EPP

Moulded plastic toys

Moulded toy components developed with an intended age group, use pattern and safety evaluation in mind.

Material options: EPP / EPS

Childhood accessories

Foam and polymer accessories for children's product concepts, specified around their actual use and contact surfaces.

Material options: EPP / EPS

INDUSTRY 01 | 14 PRODUCT FAMILIES

Sports, Leisure & Early Childhood — continued

Modular foam play forms

Connectable play shapes for age-specific handling and construction.

Material options: EPP

Educational toy shapes

Moulded forms developed around learning and play activities.

Material options: EPP

Child-seat foam cores

Foam parts evaluated within child-seat and booster systems.

Material options: EPP

Nursery foam storage forms

Storage geometry for supervised childcare and nursery settings.

Material options: EPP

Movement-course play blocks

Play forms for supervised balance and movement activities.

Material options: EPP

INDUSTRY 02 | 13 PRODUCT FAMILIES

Logistics & Material Handling

Protect, organise and move products through storage, transport and return journeys.

Specification focus: Consider stacking loads, usable volume, cleaning, return journeys and how operators lift or access the packed product.

Plastic pallets

Polymer pallet formats for moving unit loads, selected around handling equipment and the actual goods distribution system.

Material options: EPP / EPS / PP

Industrial storage bins

Containers for organising components, work-in-progress and returnable material flows.

Material options: EPP / EPS / PP

Custom cushioning foams

Product-specific foam inserts and support forms that control movement within the transport pack.

Material options: EPP / EPS / PP

Polystyrene insulated boxes

Cellular foam containers for temperature-sensitive shipments, specified as part of a complete pack configuration.

Material options: EPP / EPS / PP

Stackable delivery boxes

Insulated delivery formats designed around filling, stacking and the last metres of the delivery journey.

Material options: EPP

Catering transport boxes

Insulated containers for moving prepared meals and catering loads through a defined service route.

Material options: EPP / EPS / PP

Foldable insulated boxes

Collapsible insulated formats for applications where storage and return volume are important considerations.

Material options: EPP / EPS / PP

Reusable industrial containers

Returnable containers for protected component handling, collection and storage across repeated journeys.

Material options: EPP

Reusable industrial coolers

Returnable insulated transport formats for repeated temperature-sensitive distribution cycles.

Material options: EPP

INDUSTRY 02 | 13 PRODUCT FAMILIES

Logistics & Material Handling — continued

Lightweight foam pallets

Foam pallet forms for lightweight unit-load handling and packaging development.

Material options: EPS

EPS transport boxes

Lightweight EPS container formats for developing protective or insulated transport packs.

Material options: EPS

Industrial pallet boxes

Container formats for palletized storage and handling.

Material options: EPP

Adhesive foam support pads

Cellular support pads for locating and supporting products inside a protective packaging unit.

Material options: EPP / EPS / PE

INDUSTRY 03 | 10 PRODUCT FAMILIES

HVAC

Bring insulation, air management and assembly into a more considered component design.

Specification focus: Airflow geometry, thermal interfaces, acoustic behaviour and access for servicing belong in the same design brief.

Custom HVAC components

Shaped components that bring air-channel geometry, insulation and equipment interfaces into one coordinated design.

Material options: EPP / EPS

Moulded air ducts

Lightweight shaped air passages for connecting sections of an HVAC assembly with a defined airflow path.

Material options: EPP / EPS

HVAC insulation housings

Fitted insulation shells around equipment sections, pipework interfaces and thermally sensitive assemblies.

Material options: EPP / EPS

Heat-pump covers

Cover forms fitted around heat-pump units.

Material options: EPP / EPS

Heat-pump support frames

Support frames arranged around heat-pump assemblies.

Material options: EPP / EPS

Hydraulic module supports

Support forms for hydraulic assemblies and their connections.

Material options: EPP / EPS

Water-heater insulation forms

Insulation geometry fitted around water-heating equipment.

Material options: EPP / EPS

Hydraulic manifold insulation

Fitted insulation around manifolds and heat-distribution connections.

Material options: EPP / EPS

Condensate collection trays

Collection forms integrated into cooling equipment.

Material options: Material selection required

Fan-coil structural supports

Support geometry for fan-coil and terminal assemblies.

Material options: Material selection required

INDUSTRY 04 | 10 PRODUCT FAMILIES

Automotive

Develop lightweight vehicle components and protective inserts around the complete automotive assembly.

Specification focus: Review assembly clearances, repeated loading, temperature exposure, handling and end-of-life separation.

Mobility foam components

Lightweight moulded forms for vehicle assemblies where integration, packaging space and component handling matter.

Material options: EPP / EPS

Vehicle protective inserts

Shaped inserts that hold or separate vehicle components through production handling and transport.

Material options: EPP / EPS

Vehicle floor panels

Floor forms specified around installed space and loading.

Material options: EPP

Vehicle seat foam components

Foam forms evaluated within vehicle seating assemblies.

Material options: EPP

Vehicle storage organizers

Fitted storage layouts for tools and loose equipment.

Material options: EPP

Battery protective inserts

Protective components specified around battery integration.

Material options: EPP

Vehicle insulating panels

Panels evaluated within vehicle insulation assemblies.

Material options: EPP

Vehicle energy absorbers

Foam components evaluated within impact-related vehicle assemblies.

Material options: EPP / EPS

Vehicle acoustic inserts

Shaped inserts specified around cabin noise and vibration.

Material options: EPP / EPS

Vehicle parts shuttle trays

Returnable layouts for component transport and repeated handling.

Material options: Material selection required

MATERIAL DECISIONS

EPP and EPS: compare the application

Consideration	EPP	EPS
Base material [2]	Expanded polypropylene.	Expanded polystyrene.
Typical selection priority [2]	Resilient components and repeated handling.	Lightweight, economical insulation.
Thermal role [2]	Insulation with application-specific thickness and grade.	Insulation with application-specific thickness and grade.
Cost decision [2]	Compare tooling, density and service life.	Compare tooling, density and service life.
End of life [2]	Confirm a suitable EPP collection and recycling stream.	Confirm a suitable EPS collection and recycling stream.

Common specification questions

Is EPP the same as solid polypropylene?

It uses polypropylene in an expanded cellular form. Solid PP mouldings and EPP foam components therefore need different geometry and performance specifications. [1]

How should I choose a density?

Start with the required load response and geometry. Distinguish loose-bead bulk density from finished moulded density; ask for relevant curves and tolerances. [5]

Can a part take repeated impacts?

Resilience supports repeated use, but damage and permanent deformation remain possible. Agree realistic impact conditions and inspection or replacement criteria. [4]

Does low water absorption mean waterproof?

Low material uptake does not establish a watertight container. Validate closures, joints and drainage against the intended exposure. [1]

QUALIFICATIONS & SERVICE LIFE

Ask for the evidence that matters

Is EPP fireproof?

No blanket fire rating applies. Flame-retardant grades exist; request test evidence for the chosen grade, density, thickness and intended component. [3]

Can EPP contact food?

Some grades have food-contact approval. Request current documentation for the exact material and intended conditions; the finished article may require migration testing. [3]

Which temperature range can I specify?

Use the selected grade data and validate exposure duration, load and assembly fit. A generic foam temperature claim is insufficient for equipment design. [3]

Is every EPP part recycled or recyclable locally?

Recycled content varies by grade. Recyclability also needs a collection route that accepts the finished part, including its inserts, labels and contamination. [3]

How accurate can a moulded part be?

Shrinkage and tolerances depend on grade, density, tooling, geometry and processing. Agree critical dimensions and measurement conditions with the moulder. [5]

What should be tested before approval?

Test representative parts in the intended assembly. Agree load, fit and repeated-use checks, then add application-specific thermal, acoustic or qualification tests. [5]

Start a project conversation

Send the drawing revision, target quantities, operating conditions and acceptance criteria to sales@parknonwoven.com. Confirm the final material, geometry and validation plan for the specific application.

PRIMARY REFERENCES

Material and design sources

Numbered references identify the producer guidance behind the material guide. These sources describe material behaviour and selection considerations; they do not certify a PARK product or assembly.

[1] BEWI

EPP material properties and production

[Open primary source](#)

[2] Knauf Industries

EPP and EPS for technical HVAC components

[Open primary source](#)

[3] JSP

ARPRO grade selection sheet

[Open primary source](#)

[4] Knauf Industries

EPP material, moulding and applications

[Open primary source](#)

[5] JSP

ARPRO moulded-part design principles

[Open primary source](#)

Use current grade documentation and representative component tests when making a specification. Performance, approvals and available recycling routes depend on the selected material and finished design.

This edition follows the current online catalogue: 47 product families in 4 industries.